

intermediate-elective

Linda Herrera

2026-05-10

Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizations and coloring
library(RColorBrewer)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")
```

```
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

# vegetation information
veg <- read_csv(here("data", "veg.csv"))

# vegetation meta data information
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Problems

1. Explain your inspiration

Which visualization you chose: The visualization I chose for inspiration is a circular bar chart made by Cristina Cametti using the IKEA TidyTuesday dataset, which shows the most common IKEA designers grouped by furniture category.

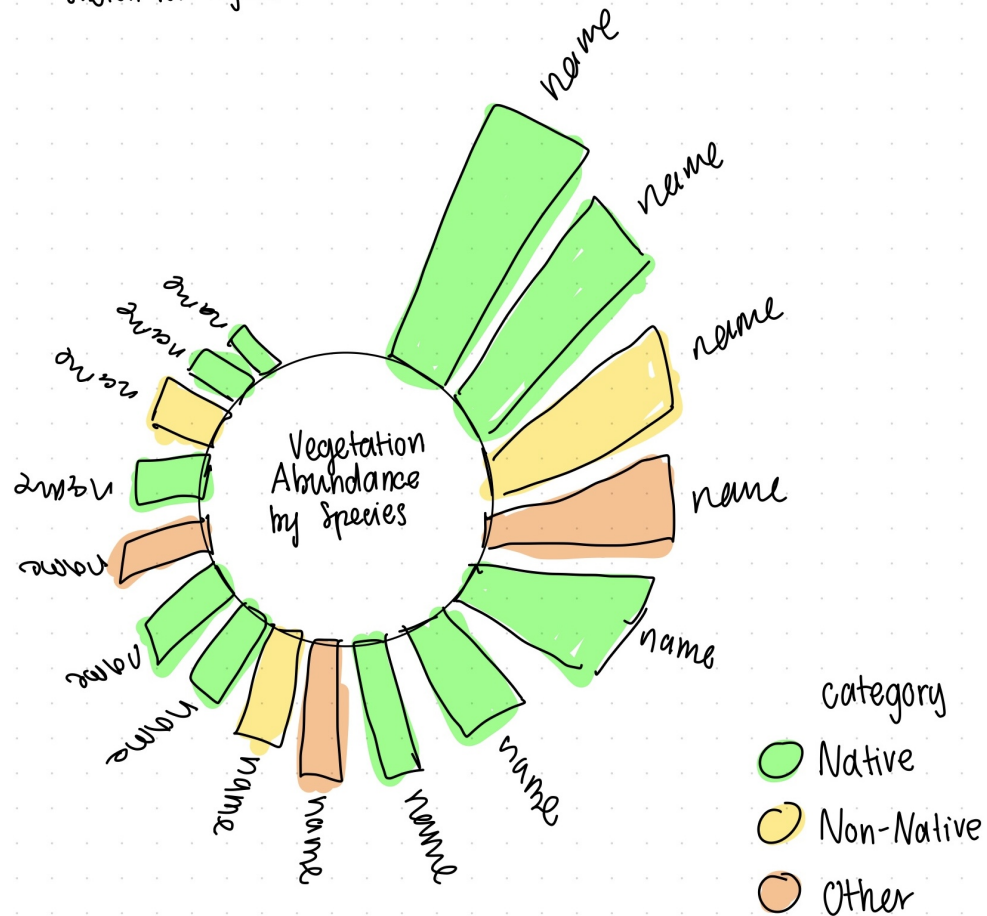
Why it makes sense for your dataset: This works well for my vegetation data because there are many plant species across different cover categories, and the circular layout is a clean way to display a lot of species at once without the labels overlapping like they would in a regular bar chart.

Which variables you're using: The three variables shown are plant species (`psoc`), cover category (`cover_category`), and total percent cover, which is the sum of `percent_cover` across all sampling locations.

How variables map onto visual components: Plant species map onto the individual bars with their names as labels around the outside of the circle, total percent cover maps onto bar height, and cover category maps onto bar color, which mirrors exactly how the IKEA inspiration mapped designer names, item counts, and furniture category onto the same visual components.

2. Plan your figure

Sketch for figure



3. Code your figure

```
# wrangling: sum percent cover by species and cover category
data <- veg |> # starting with veg data frame
  group_by(cover_category, psoc) |>
  summarise(n = sum(percent_cover, na.rm = TRUE)) |>
  ungroup() |>
  # removing bare ground, thatch, and NA cover categories
```

```

filter(!psoc == "Bare Ground") |> filter(!cover_category == "BARE GROUND") |>
filter(!cover_category == "THATCH") |> filter(!cover_category == "NA") |>
filter(!is.na(cover_category)) |>
# replacing underscores with spaces in species names
mutate(psoc = str_replace_all(psoc, "_", " ")) |>
# renaming cover categories for legend
mutate(cover_category = recode(cover_category,
                              "NATIVE COVER" = "Native Cover",
                              "NON-NATIVE COVER" = "Non-Native Cover",
                              "OTHER COVER" = "Other Cover"))

# keep only species with high abundance to avoid overcrowding visual
dt <- data |>
  arrange(desc(n)) |>
  filter(n > 500) # set the threshold

# converting cover category and species to factors
dt <- mutate_at(dt, vars(psoc), as.factor)

# setting factor levels explicitly to exclude NA from legend
dt$cover_category <- factor(dt$cover_category,
                           levels = c("Native Cover", "Non-Native Cover", "Other Cover"))

# adding empty bars at the end for spacing
empty_bar <- 2
to_add <- matrix(NA, empty_bar, ncol(dt))
colnames(to_add) <- colnames(dt)
dt <- rbind(dt, to_add)

# creating an id column for polar coordinates
dt$id <- seq(1, nrow(dt))

# calculating label angles and alignment for readability
label_data <- dt
number_of_bar <- nrow(label_data)
angle <- 90 - 360 * (label_data$id - 0.5) / number_of_bar
label_data$hjust <- ifelse(angle < -90, 1, 0)
label_data$angle <- ifelse(angle < -90, angle + 180, angle)

# setting color palette for cover categories
pal <- brewer.pal(3, "Dark2")

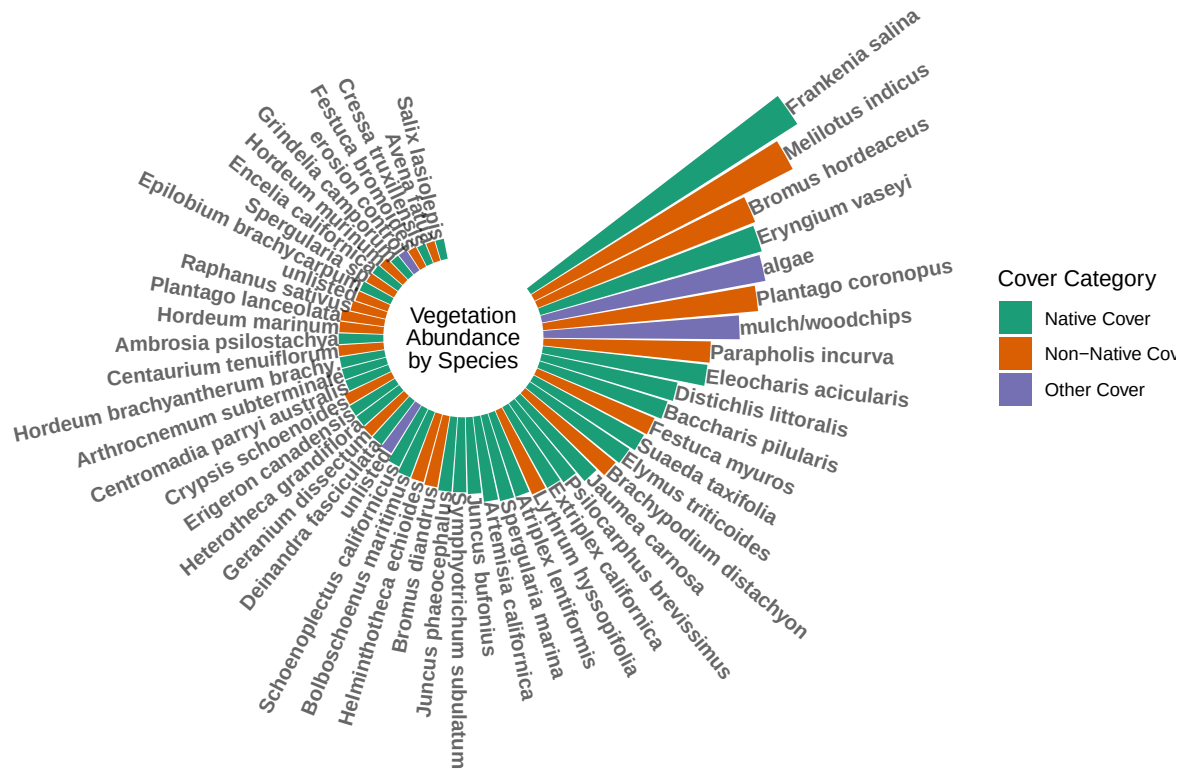
```

```

# plotting circular bar chart
plot <- ggplot(dt) +
  # bars filled by cover category
  geom_bar(aes(x = as.factor(id), y = n, fill = cover_category), stat = "identity") +
  # applying color palette, removing NA from legend, and setting legend title
  scale_fill_manual(values = pal,
                    na.translate = FALSE,
                    name = "Cover Category") +
  # setting y limits to create space in the center and top
  ylim(-2000, 8000) +
  # classic theme
  theme_classic() +
  # removing axes and grid lines
  theme(
    axis.text = element_blank(),
    axis.title = element_blank(),
    panel.grid = element_blank(),
    plot.margin = unit(rep(-1, 4), "cm")) +
  # center title
  annotate(geom = "text",
    x = 0, y = -2000,
    hjust = .5, vjust = 0.5,
    label = "Vegetation\nAbundance\nby Species", # added extra line break
    size = 4,
    lineheight = .8,
    color = "black") +
  # converting to polar/circular layout
  coord_polar(start = 0) +
  # adding species name labels around the outside
  geom_text(data = label_data,
    aes(x = id, y = n + 1, label = psoc, hjust = hjust),
    color = "black", fontface = "bold",
    alpha = 0.6, size = 3.4,
    angle = label_data$angle,
    inherit.aes = FALSE)

# displaying plot
plot

```



4. Describe your process